Machine Data and Learning

Python ML Yuxi Hayden Liu
Artificial Intelligence: A Modern Approach Stuart Russell and Peter Norvig

AI:- That which mimics human intelligence in machines
The science and engineering of making machines (like comps. robots) intelligent & enabling them to take actions and decisions that humans do

Machine:- Physical system that uses power to apply forces and control movement to perform action (to make life easier)

Complex machines are combination of 2 or more simple machines

AI has been proliferating healthcare into etc

AI for social good:- Computer Vision, help farmers, supply chain management, Public & wildlife safety, Automated Event Discovery, Border Patrol, etc.

What is intelligence?
The ability to understand, learn and think.

The capacity - Wiki: defn peeps.

We think, learn from observation, and make decisions

But what about machines?

Why is he such an arse? :(

Two dimensions to AI

Humaneness vs Rationality

Act vs Thinking

Turing Test Human interpreter cannot distinguish b/w comps & humans after ~~processing~~ posing some written questions.
C needs NLP, Knowledge rep, Automated reasoning, ML, CV & Robotics
Turing test involves interaction with objects and people

Think Humanly :- Cognitive modelling

Act Rationally :- Agent based modelling, Agents should operate autonomously, perceive the environment, adapt to change

Think Rationally :- Right thinking, Logic based approach, need to involve probability to model uncertain things

What is ML?
Construction of algos that can learn from, and make predictions on data.

data :: features (feature modelling?) [CL2 saving my arse lol]

Supervised ML :- Classification, Regression [Bayesian modeling]

Unsupervised ML :- Clustering, Anomaly detection features for differentiating b/w cricket and tennis ball would be wt and diameter [plot on graph, CL vectors]

Perceptron :- Observe feats, apply activation fn and produce output

Linearity in features makes computation easier

If x_1 = width and x_2 = wt

$f_n = w_0 + w_1 x_1 + w_2 x_2$

w_0 ↓ bias → used to distinguish b/w the 2 balls easily

Neural Networks and Deep Learning

Replicate neurons in human brain

Where does AI stand today?

Significant progress in last three decades in CV, NLP, speech recog.

Far from ~~[illegible]~~ mimicking humans perfectly

Human intelligence is not purely ~~[illegible]~~ reflex mechanisms but is dependent on reasoning that operate on the internal representations of knowledge

AI is about building machines and sys that can reason, learn & solve problems similar to humans. logical reasoning is key to allowing sys to decide based on given conditions & facts

Knowledge base :- Set of sentences mapped in knowledge representation language.
↳ Expressing knowledge explicitly in a computer-tractable way. (logic? wot?)

Logic :- A way of thinking or explaining, sensible reasons for doing smth, sci of thinking about or explaining the reasons for smth using formal methods.

Basis of all mathematical and automated reasoning

Formal vs Informal Logic

Formal	Informal
Formal Lang	Natural Lang
Study of logical Truth	Study of informal fallacies, crit. thinking
Symbolic / Mathematical Logic	

Natural Lang exhibit ambiguity
Makes ~~reasoning~~ reasoning difficult (ambiguity

Formal Lang
- promotes rigour, reduces possibility of human error
- helps reduce implicit / unstated assumptions by removing familiarity with subject matter
- helps achieve generality due to possibility of finding alt interpretation for sentences & args.

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Formal Logic :- Uses syllogisms to make inferences & examines how conclusion follow from premises based on the structure of args
Mathematical Logic :- Uses math symbols to prove theoretical args
Symbolic Logic :- Uses symbols to accurately map out valid & invalid args
Propositional Logic

4 Imp. cause of it being
(i) Core of AI :- Possibility of automating reasoning draw inferences from knowledge
Answer queries
discover facts that follow from the knowledge base, decide what to do
Essential for knowledge rep, reasoning, decision-making process

Entailment :- Deducing a statement from other statements

T → T
F → T/F
T/F ← T
F ← F

(A) ∀ Humans have 2 eyes
(B) Sujit is human

(P) Sujit has 2 eyes

(C) ∀ Humans have 4 eyes (Q) Sujit has 4 eyes
(D) Sujit is human

Pack my bag with five dozen liquor jugs

Pack my bag with five dozen liquor jugs

Fallacy :- Conclusion may be right, but reasoning is faulty

Propositional Logic :- Deals with propositions (statements) which are T/F . Propositional calculus . Zero order logic Foundation of higher order logics

Order of Logic Degree of quantification that can be performed over sets .

(i) Ist order :- Quantifies only over individuals . Predicate logic /calculus
(ii) IInd order :- Over sets /predicates (acc to Prats :3)
(iii) IIIrd order :- Over sets of sets
(iv) Higher orders :- Quantification over arbitrarily deep nesting

Propositions are declarative statements

(i) Atomic :- Simple, indivisible, cannot be chunked further
Represents basic fax or condition
"The door is closed"
(ii) Compound :- Atomic prop. combined using logical connectives (AND, OR, NOT)
"The door is closed AND the heater is on"

Connectors :-

(i) AND ($\wedge$)
(ii) OR ($\vee$)
(iii) NOT ($\neg$)
(iv) IMPLIES ($\rightarrow$) $S = \neg$ if $P \rightarrow Q$
(v) IFF ($\leftrightarrow$) $P \leftrightarrow Q = \neg P$ XOR Q $= P \rightarrow Q \wedge Q \rightarrow P$

$P \rightarrow Q$:- P is necessary for Q
Q is sufficient for P

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$P \leftrightarrow Q$:- P & Q are equivalent
Q & P are necessary & sufficient

Truth Table :- break down of all possible truth values returned by a logical expr.

P	Q	$P \to Q$	$\neg P \vee Q$	$P \leftrightarrow Q$	$\neg P \oplus Q$
0	0	1	1	1	1
0	1	1	1	0	0
1	0	0	0	0	0
1	1	1	1	1	1

P & Q can be compound propositions

Exercise:- A & B = T , X & Y = F

(i) $\neg(A \vee X)$ F
(ii) $A \vee (X \wedge Y)$ T
(iii) $A \wedge (X \vee (B \wedge Y))$
(iv) $((A \wedge X) \vee \neg B) \wedge \neg((A \vee X) \vee \neg B)$
(v) $(X \wedge Y) \wedge (\neg A \vee X)$
(vi) $((X \wedge Y) \to A) \to (X \to (Y \to A))$

Terminology

Tautology:- Invariably True for all assignments → Valid sentences (Null sentences?)

SAT:- A sentence is satisfiable iff there is some assignment of values that results in True
A sentence is UNSAT if it is $\neg$ SAT

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Left hand

Towards Formal Proof

$\alpha \Rightarrow \beta$ when all the formulae in set α are True, β is True
Semantic notion, concerns the notion of truth (?)
(Grammar died)

Construct a Truth Table for α, β
$\alpha \Rightarrow \beta$ if $\exists$ row where all formulae in α are true, β is also True

Modus Ponens :-

$\alpha: P \wedge (P \to Q)$; $\beta: Q$

Write TT for $P, P \to Q$

Therefore $P, P \to Q \Rightarrow Q$

Intends to formally capture the notion of proof that is commonly applied in other fields

Proof:- of a formula from a set of premises is a seq of steps in which any step of the proof is

i) an axiom
ii) a formula deduced from the prev step using some rule of inference
iii)

The last step of the proof should deduce the formula we wish to prove.

We say that S follows from (premises) α to denote that the

Towards Formal

set of formulae α "proves" the formula s

Soundness:- A logic is sound if it preserves truth (i.e. if a set of premises are all true, any conclusion drawn from those premises must also be true)

Completeness:-

(i) Semantic:- A logic is complete if it is capable of proving any valid consequence

(ii) Syntactic:- Completeness requires that every sentence, or its negation is provable (i.e. a theorem)

Decidability:- A logic is decidable if $\exists$ a mechanical procedure (computer program) to prove any given consequence. (or) $\exists$ a decision procedure that is finite for theorem

No logical system is syntactically complete

Zero Order Logic is semantically complete, so is FOL

ZOL is decidable, FOL is not

ZOL inference rules

(i) Modus Ponens $P, P \rightarrow Q \Rightarrow Q$

(ii) Modus Tollens (Contraposition) $P \rightarrow Q, \neg Q \Rightarrow \neg P$

(iii) Hypothetical Syllogism $P \rightarrow Q, Q \rightarrow R \Rightarrow P \rightarrow R$

(iv) And Elimination : $\bigwedge_{i=1}^{n} P_i \Rightarrow P_i$

(v) And Introduction $P_1, P_2, \cdots P_n \Rightarrow \bigwedge_{i=1}^{n} P_i$

(vi) Or Introduction $P_i \Rightarrow \bigvee_{i=1}^{n} P_i$

(vii) ~~Dot~~ Double Negation Elimination $\neg\neg P \Rightarrow P$

(viii) Unit Resolution $P \vee Q, \neg Q \Rightarrow P$

(ix) Resolution $P \vee Q, \neg Q \vee R \Rightarrow P \vee R$

Example :-

	α	β
1)	$A \vee (B \to D)$	$B \to E$
2)	$\neg C \to (D \to E)$	
3)	$A \to C$	
4)	$\neg C$	

Proof

5	$D \to E$	2,4	(Modus Ponens)
6	$\neg A$	3,4	(Modus Tollens)
7	$B \to D$	1,5	(Unit Resolution)
8	$B \to E$	5,7	(Hypothetical Syllogism)

Q) ~~Construct formal proof for :-~~
~~If the investigation~~

$S \to G$
$\neg S \to E$
$\neg G \to E$

Conjunctive Normal Form (CNF)

A literal is a propositional letter or the negation of a propositional letter

A clause is a disjunction of literals

CNF → Conjunction of clauses
DNF → Disjunction of ~~clauses~~ conjunction of literals

Every propositional logic-formula can be converted to CNF and DNF

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Converting to CNF

(i) Eliminate $\leftrightarrow$ rewriting $P \leftrightarrow Q$ as $(P \to Q) \land (Q \to P)$
(ii) Eliminate $\to$ as $P \to Q \Rightarrow \neg P \lor Q$
(iii) Use ~~push $\neg$~~ De Morgan's laws to push $\neg$ inwards
(iv) Rewrite $\neg (P \land Q)$ as $\neg P \lor \neg Q$
(v) Rewrite $\neg (P \lor Q)$ as $\neg P \land \neg Q$
(vi) Eliminate double negation $\neg\neg P \Rightarrow P$
(vii) Use distributive laws to get CNF
(viii) Rewrite $(P \land Q) \lor R$ as $(P \lor R) \land (Q \lor R)$
(ix) Rewrite $(P \lor Q) \land R$ as $(P \land R) \lor (Q \land R)$

Q) Convert $\neg (P \to (Q \land R))$

$\neg (\neg P \lor (Q \land R))$
$\neg\neg P \land \neg (Q \land R)$
$P \land (\neg Q \lor \neg R)$

Parchuri Time! (Hopefully don't regret this)

- Bro's sharing slides! Goat!

Only taking down topic names, ought to refer to the textbooks

ML :- Scientific study of algos & statistical models that computer systems use
1) to perform a specific task effectively sans explicit instr.
2) rely on patterns & inferences instead

Involves building a math model based on sample data

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$$E[(y-\hat{f}(x))^2] \Rightarrow E[(f+\varepsilon-\hat{f})^2] = E[(f-\hat{f})^2 + \varepsilon^2 + 2\varepsilon(f-\hat{f})]$$

$$= E[(f-\hat{f})^2] + E[\varepsilon^2] + 2E[\varepsilon(f-\hat{f})]$$

$$= E[(f-E[\hat{f}]+E[\hat{f}]-\hat{f})^2] + \sigma^2 + 0 \quad \text{[Independence of noise]}$$

$$= E[(f-E[\hat{f}])^2 + (E[\hat{f}]-\hat{f})^2 + 2(f-E[\hat{f}])(E[\hat{f}]-\hat{f})] + \sigma^2$$

$$= E[(f-E[\hat{f}])^2] + E[(E[\hat{f}]-\hat{f})^2] + 2(f-E[\hat{f}])E[E[\hat{f}]-\hat{f}] + \sigma^2$$

$$= (f-E[\hat{f}])^2 + \text{Var} + 0 + \sigma^2 = \text{Bias}^2 + \text{Var} + \sigma^2 \quad //$$

Q) Let $\hat{f}(x) = \theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3$

Minimise MSE

no of training samples

$$J(\theta) = \frac{1}{m} \times \min \sum_{i=1}^{m} \left(\hat{f}_\theta(x^{(i)}) - y^{(i)}\right)^2$$

We wish to penalise x^2 & x^3

Add $+1000\,\theta_2 + 1000\,\theta_3$

Back To Sujit!

Data Science:-

↳ clean → visualise (or visdualise acc to Sujit) → analyse → predict ↵

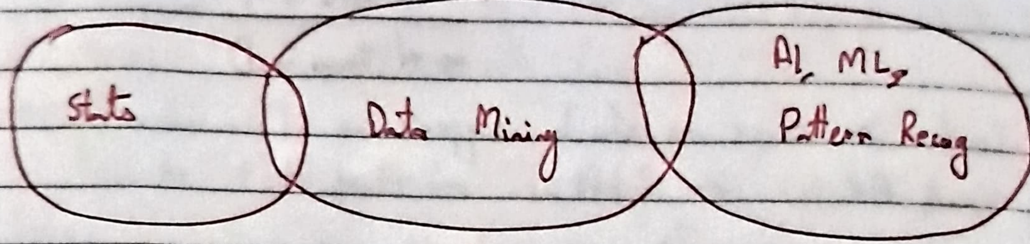
Interdisciplinary academic field that uses stats, scientific computing, scientific methods, processing, ... (ref wiki for defn)

Business
Govt
Research Labs
Internet

→ Data $\xrightarrow{\text{Mine / explore}}$ Patterns → Decision Making → Feedback to source

Data Mining is the process of discovering valuable insights, patterns and information from vast datasets by employing various techniques, algorithms & tools

It is a technology that blends traditional data analysis methods with sophisticated algos for processing large amts of data



Data base Tech, $\|^{lel}$ comp, Distr. Comp

Patterns:-

- Association:- Coffee buyers also buy sugar.
- Clustering:- Segregation & Segmentation of data.
- Classification:- Bank customer applies for loan, Will they default?

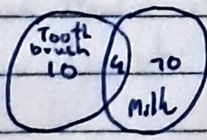
Association Rules:-

- Antecedent & Consequent
- AR: $X \rightarrow Y$

$$\{x_1 \cdots x_n\} \rightarrow \{y_1 \cdots y_k\}$$

- itemset is the list of all the items in the antecedent & the consequent i.e. $X \cup Y$

For ex:-



Assoc. Rule

A.R:- Toothbrush $\rightarrow$ Milk

Support = $\frac{4}{100}$

Conf = $\frac{4}{14}$

$$\text{Support} \Leftarrow \frac{\text{no. of transactions in } X \cap Y}{\text{no. of transactions in } D} = P(X \cap Y)$$

$$\text{Confidence} \Leftarrow \frac{\text{no of transactions in } X \cap Y}{\text{no of transactions in } X}$$

Ex:- Let $D = 200$

(i) $X =$ Sugar ; $|X| = 100$; $Y =$ Milk ; $|Y| = 160$; $|X \cap Y| = 70$

Support = 0.35 ; Conf = 0.7

(ii) $X =$ Coffee ; $|X| = 50$; $|X \cap Y| = 38$

Support = 0.19 ; Conf = 0.76

(iii) $X =$ Bread ; $|X| = 20$; $Y =$ ~~Bread~~ Butter ; $|Y| = 26$; $|X \wedge Y| = 15$

Support = 0.075 ; Conf = 0.75

Lift is symmetric across X & Y

$$\text{Lift}(AR) = \frac{P(Y|X)}{P(Y)} = \frac{\dfrac{\text{no of trans in } X \cap Y}{\text{no of trans in } X}}{\dfrac{\text{no of trans in } Y}{\text{no of trans in } D}} = \frac{|X \cap Y| \times |D|}{|X||Y|}$$

Conf $\geq$ 0.05 → check support → if considerable
good AR ← if lift > 1 ← check lift ↵

$$\overset{n}{X} \longrightarrow \overset{n-1}{Y} \quad \rightarrow \text{no of AR} = \sum_{k=1}^{n} {}^{n}C_k \times (2^{n-k} - 1)$$

$X \subseteq [n]$

$Y \subseteq [n]$ $\quad = 3^n - 2^{n+1} + 1$ By inclusion exclusion

↳ each element has 3 choices.

But we still have to explore 2^{101} itemsets if they meet reasonable support

$$|X \cup Y| < \text{threshold} \rightarrow |X \cup Y \cup Z| < \text{threshold}$$

If any set has a bad support, all its supersets also have a bad support.

↳ Apriori algorithm:

Itemset can be frequent only if all its subsets are frequent

No superset can be frequent if the itemset is not frequent

Example:

tx.id		Trial 1	Trial 2	Trial 3
1	{1, 2, 3, 4}	1 = 3	1,2 = 3	
2	{1, 2, 4}	2 = 5	~~1,3 = 1~~	
3	{1, 2}	3 = 4	~~1,4 = 2~~	
4	{2, 3, 4}	4 = 5	2, 3 = 3	
5	{2, 3}		2, 4 = 4	
6	{3, 4}		3, 4 = 3	
7	{2, 4}			

$\{\{\emptyset\}\}$

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Pseudo code:-

1) Create 1-sized frequent itemset lists that meet a threshold support
2) Expand itemsets list by combining overlapping sets to size+1
4) Remove infrequent itemsets from the list
~~Repeat~~
3) Prune the expanded lists using the apriori property
Loop 2, 3, 4 until no expansion is possible

Algorithm:-

```
Apriori (T, ε):
    L1 = {large singleton itemsets}
    k = 2
    while L(k-1) is not empty:
        Ck = generate_candidates (L(k-1), k)
        for t in T:
            Dt = {c if c ⊆ t for c in Ck}
            for c in Dt
                count [c] += 1
        Lk = {c if count[c] ≥ ε for c in Ck}
        k += 1
    return Union {Lk} for all k

generate_candidates (L, k):
    result = set()
    for all p, q in zip(L, L):
        if p and q differ by one element exactly:
            c = p ∪ q
            for u ⊆ c:
                if u ∈ L and |u| = k-1:
                    result.add (c)
    return result
```

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Hierarchical and Cyclic association rules also exist

Eclat Algorithm :- Eq. class transformation ; a DFS strategy. Works best when the dataset is dense.

F-P Growth :- Freq. Pattern Growth, a compact data structure = FP-tree to compress the data set. Works better than the rest when the data is dense and large.

Classification :-

We collect different measurements / facts / features

$x = \{x_1 \ldots x_d\}$

set of classes $y = \underbrace{\{1, 2, \ldots k\} = [k]}$ (set of k)

If $k = 2$ ⇐ binary classification
else ⇐ multiclass

How do we measure the goodness of a classifier?

Accuracy: $\mathbb{P}(\hat{y}_x = y_x)$

$\hat{y}_x$ ↳ predicted value

y_x → gold

$$= \frac{\text{no of pts classified correctly}}{\text{no of data pts}} = \frac{TP + TN}{TP + TN + FP + FN}$$

But fails in case of skewed data.
So we assign loss wts to fix this issue.

But we usually use Recall & Precision

		P	N ⇐ Predicted
True ⇒	P	TP	FN
	N	FP	TN

Recall = True Positive Rate = $\frac{TP}{TP+FN}$

Precision = Positive Prediction Value = $\frac{TP}{TP+FP}$

False Negative Rate = $\frac{FN}{TP+FN}$

$FPR = \frac{FP}{FN+TN}$ $\quad$ $TNR = \frac{TN}{TN+FP}$

F_1 Score $= \dfrac{2}{\frac{1}{\text{Recall}} + \frac{1}{\text{Precision}}}$

Bayes Classifier :-

$Y \in \{1 \dots k\}$; Prior = $\mathbb{P}(Y=k)$ = probability that a datapoint belongs to class k ; $\mathbb{P}(Y=k \mid X=x)$ = Posterior. We assign the MAP estimate: Max A Posteriori = $\underset{k}{\operatorname{argmax}}\ \mathbb{P}(Y=k \mid X=x) = \dfrac{\overbrace{\mathbb{P}(X=x \mid Y=k)}^{\text{Likelihood}} \times \overbrace{\mathbb{P}(Y=k)}^{\text{Prior}}}{\mathbb{P}(X=x)}$ → evidence or normalising const

Naive Bayes Classifier :- Assumption of Conditional Independence